

0 clubs; 2.5 hours (on examples)

Problem 1 (2022 12A/10)

Clearly, the numbers $n + 1, \dots, 2n$ must be each be paired with distinct numbers from $1, \dots, n$. Notice that n can be paired with $2n$, $n - 1$ can be paired with $2n - 2, 2n - 1$, $2n, n - 2$ can be paired with $2n - 4, 2n - 3, 2n - 2, 2n - 1, 2n$.

Starting with n and going down to 1, the options each number can be paired can only expand. Therefore, we use construction to get,

$$= \begin{cases} \left(\frac{n}{2}!\right)^2 & \text{if } n \text{ is even} \\ \left(\frac{n-1}{2}!\right)\left(\frac{n+1}{2}!\right) & \text{if } n \text{ is odd} \end{cases}$$

Problem 2 (2022 12A/19)

We mark the locations of the cards we pick up on the first pass as A and that of the second pass as B. There is a bijection between all possible passes and all $2^{13} - 14$ ways to write 13 As and Bs excluding AA...AA, AA...AB, AA...ABB, ..., AB...BB, BB...BB which cannot be achieved.

The answer is hence $8192 - 14 = 8178$.

Problem 3 (ARML 2023 T-6)

We casework on n how many numbers we have.

Note that since 17 is prime, for $1 \leq n \leq 16$, it'd be for us to get the same thing after $k < n$ cyclic rotations since that'd suggest $\frac{n}{k} \mid 17$, which is impossible. Therefore, we can just count how many ways to write 17 as the sum of n ordered positive numbers and divide by n . By stars and bars, our sum is,

$$= \sum_{n=1}^{16} \frac{1}{n} \binom{16}{n-1} = \frac{1}{17} \sum_{n=1}^{16} \binom{17}{n} = \frac{2^{17} - 2}{17} = 7710$$

For $n = 17$, we have 1 case.

In total, there are 7711 ways to do this.

7 clubs; 4 hours

Problem 5 (Z938DBE5)

Color the grid as a checkboard. The white squares point to black squares and vice versa, so we independently count how many ways we can direct the white squares to black squares such that the indegree of each black square is exactly one and that of black squares to white squares.

B 1	2	B 3	4
5	B 6	7	B 8
B 9	10	B 11	12

We casework on where cells 1 and 9 can point.

Case 1: Both 1 and 9 points right.

Cell 6 must point left. The remaining cells can behave in 3 ways.

Case 2: Exactly one of them points right.

WLOG, let 1 point right and 9 up.

WLOG, let 8 points to 12, meaning 3 must point to 4 and we'd have two cases for how 6 and 11 behave.
In total, there are $2 \cdot 2 \cdot 2 = 8$ cases here.

In total, we'd have 11 cases. The answer is $11^2 = 121$.

Problem 6 (HMMT 2018 C6)

We let x_n be the probability of Rachel reaching $(0, 0)$ if she is at (n, n) .

$$x_n = 1 - \left(\frac{1}{2}\right)^n$$

We prove this using induction,

Base case: $n = 0$, the two already meet.

Inductive step: $n = k$ to $n = k + 1$.

Note that, by symmetry, Rachel and Sarah have the same probability of winning no matter what first move they make, so we just consider that they have equal probability of moving along either axis.

Rachel and Sarah can only meet at one of the points,

$$(0, k + 1), (1, k), (2, k - 1), \dots, (k - 1, 2), (k, 1), (k + 1, 0)$$

If Rachel and Sarah both move along the x-axis, Rachel can just spam moving x-axis until she gets to $(0, k + 1)$. Same thing if they both move along the y-axis.

If they move along different axis, we'd get the x_k case.

Therefore, the probability is, $x_{k+1} = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \left(1 - \left(\frac{1}{2}\right)^k\right) = 1 - \left(\frac{1}{2}\right)^{k+1}$.

Conclusion: By mathematical induction, we are done.

We now let y_n be the probability of Rachel reaching the origin if she is at $(n, n + 2)$.

We claim that,

$$y_n = 1 - \left(\frac{1}{2}\right)^n$$

We prove this using induction.

Base case: $n = 0$, the two must meet on next step.

Inductive step: $n = k - 1$ to $n = k$,

Rachel and Sarah may only meet at one of,

$$(0, n + 1), (1, n), (2, n - 1), \dots, (n, 1)$$

If Sarah moves right, no matter what Rachel's first move is, Rachel can just proceed to $(0, n + 1)$ and be safe.

If Sarah moves up and Rachel moves down, we'd have $x_n = 1 - \left(\frac{1}{2}\right)^n$ probability of Rachel winning.

If Sarah moves up and Rachel left, we'd have $y_{n-1} = 1 - \left(\frac{1}{2}\right)^{n-1}$ probability of Rachel winning.

Sarah would have to move up to not immediately lose. Rachel, knowing that, would move down to have the best chances of winning.

Conclusion: By mathematical induction, we're done.

The answer is $y_6 = 1 - \left(\frac{1}{2}\right)^6 = \frac{63}{64}$.

Problem 8 (2022 12B/17)

We prove there is a bijection between the combinations of two permutations of $\{1, 2, \dots, 6\}$, one for the rows, one for the columns, and what the matrices.

For every matrix, we'd have distinct row sums and column sums since if two distinct matrices were to have the same row sums and column sums, we'd need to permute a row, which would change the column sums.

We now show that for every row and column sum permutation, we'd have exactly one distinct matrix. To construct a matrix, we just start with,

Row \ Column sum	1	2	3	4	5	6
1	0	0	0	0	0	1
2	0	0	0	0	1	1
3	0	0	0	1	1	1
4	0	0	1	1	1	1
5	0	1	1	1	1	1
6	1	1	1	1	1	1

We can swap rows / columns around until we get our desired permutations. Finally, two identical matrices must have the same row and column sums, so we're done.

The answer is $(6!)^2 = 518400$.

14 clubs; 6 hours

Problem 11 (ECV11119)

We use complementary counting. There can only be at most 1 cycle of length more than n . We casework on the length l of said cycle,

$$\lim_{n \rightarrow \infty} (1 - p_n) = \lim_{n \rightarrow \infty} \frac{\sum_{l=n+1}^{2n} \binom{2n}{l} \cdot \frac{(l!)}{l} \cdot (2n-l)!}{(2n)!} = \lim_{n \rightarrow \infty} \sum_{l=n+1}^{2n} \frac{1}{l} = \int_n^{2n} \frac{1}{l} dl = \ln(2)$$

The answer is hence $1 - \ln(2)$.

Problem 12 (321AVOID)

We split each 321-avoiding permutation into blocks where each block starts and ends with a new running maximum (e.g., $1263745 \rightarrow (1)(2)(63)(745)$). We call the penguin start of each pair of parentheses the leader.

We show that there is a bijection to all the Catalan paths on the grid. We have the running maximum start at 0. Every step, we move right on the grid by change in running maximum and then move up by one.

This way, at point (x, y) , the running maximum is x and the number of moves we made is y . By Pigeonhole our path must stay below $y = x$.

Now to show that every Catalan path corresponds to a unique legal permutation, we notice that the leaders obviously form an increasing sequence. However, the non-leaders must also form an increasing sequence since otherwise we just add on a leader, and we'd have a 321 situation.

We keep track of the value of the previously added leader/non-leader and update by whether we move up/right.

28 clubs; 8.5 hours

Problem 16 (IMO 2011/4)

Double all the weights for convenience.

Denote the weights by their exponents, we iterate over all permutations of the weights $\{1, \dots, n\}$. Let the leaders be the same thing they were for the last problem. Since,

$$2^n > 2^{n-1} + 2^{n-2} + \dots + 2^0$$

Every time the running maximum (which starts at -1) gets updated, we must add our new weight to the left. Due to the same inequality above, at all other times, we can place our new weight on either.

The answer is hence, over all permutations S_n ,

$$\sum_{S_n} 2^{n - \# \text{Leaders}(S_n)}$$

Let a_n be the answer for a given n . We claim $a_n = (2n - 1) \cdot (2n - 3) \cdot \dots \cdot 1$. We prove this by induction,

Base case: $n = 0$, trivially 1.

Inductive step: $n = k - 1 \rightarrow n = k$

Casework on the location of the n . Let k be the number of numbers before the n .

Note that there are no leaders after the n , so we can slap on a factor of $2^{n-k-1}(n - k - 1)!$. The number of leaders before the n is unaffected, so we can also slap on a factor of a_k .

$$\begin{aligned} &= \sum_{k=0}^{n-1} \binom{n-1}{k} 2^{n-k-1} a_k (n - k - 1)! \\ &= \sum_{k=0}^{n-1} \frac{(n-1)! a_k 2^{n-k-1}}{k!} \\ &= \sum_{k=0}^{n-1} (2k-1)(2k-3) \dots (1) \cdot \frac{(2n-2)(2n-4) \dots (2)}{(2k)(2k-2) \dots (2)} \\ &= \sum_{k=0}^{n-1} (2n-2) \dots (2k+4)(2k+2) \cdot (2k-1)(2k-3) \dots (1) \\ &= (2n-2) \cdot a_{k-1} + (2n-3)(2n-5) \dots (1) = a_k \end{aligned}$$

Conclusion: We're done by math. Induction.

Problem 19 (ARML 2021 I-8)

Note that if f is not bijective, it cannot be surjective so there'd exist an n where there'd exist no $g(k) = n$, meaning $g^{g(n)}(n) = n$ is impossible. Therefore, g must form a bunch of cycles.

Claim: all the numbers in a cycle of length k are multiples of k

Proof: If we have a m in the cycle of length k where this is not met, we look at the number n also in the cycle where $g(n) = m$, we have $g^{g(n)} \neq n$, which is a contradiction.

We obviously cannot have any cycles of length 4 or more. (We do massive casework)

Problem 20 (IMO 2008/5)

The sequences of steps that are counted in N are called of the N type and same for M .

For each sequence of steps of the N type, every time we make a switch k on lamps $n + 1$ to $2n$ we can instead switch $k - n$. Therefore, each sequence of the N type can be paired to exactly one sequence M type this way.

We claim that there is the same amount of sequences of the N type paired to each sequence of the M type. For each sequence of length M let a_k be the number of times we switch k for all $1 \leq k \leq n$.

We can select $a_1 - 2k_1$ over all $2k_1 < a_1$ of the 1s to change to $n + 1$, $a_2 - 2k_2$ over all $2k_2 < a_2$, ... In total, we'd have,

$$\begin{aligned} & \left(\binom{a_1}{1} + \binom{a_1}{3} + \dots + \binom{a_1}{a_1} \right) \left(\binom{a_2}{1} + \binom{a_2}{3} + \dots + \binom{a_2}{a_2} \right) \dots \left(\binom{a_n}{1} + \binom{a_n}{3} + \dots + \binom{a_n}{a_n} \right) \\ &= \frac{2^{a_1}}{2} \cdot \frac{2^{a_2}}{2} \cdot \dots \cdot \frac{2^{a_n}}{2} = \frac{2^k}{2^n} \end{aligned}$$

Sequences of type N for each sequence of the M type.

The answer is therefore just 2^{k-n} .

38 clubs; 12.5 hours

Problem 21 (2022 10B/18)

If we have three distinct non-zero equations where no two can derive the third, we'd have only one solution. All other solutions must have infinite solutions since there'd be three variables.

Case 1: There is at least one equation where $a_k = b_k = c_k = 0$.

Complementary count to get, $= (2^3)^3 - (2^3 - 1)^3 = 169$.

Case 2: There is a repeated equation.

Considering all equations are non-zero, $= (3 \cdot 7 \cdot 6) + 7 = 133$.

Case 3: We can manipulate two equations to get the third.

Subtracting equation 1 from equation 3 to get equation 2 is equivalent to adding 1 and 2 to get 3. Since $a_k, b_k, c_k \in \{0, 1\}$, we cannot add twice an equation in any case.

WLOG, count the ways to add non-zero equation 1 to non-zero equation 2 to get equation 3 (which is obviously non-zero and distinct from 1 and 2).

If c_1, c_2, c_3 is a permutation of $(1, 1, 0)$, there'd be two cases to assign which one goes where for a total of 6 cases.

If c_1, c_2, c_3 is $(1, 1, 1)$, there'd be 6 cases.

In total, that's 12. We can also have eq. 2 + eq. 3 = eq. 1 or eq. 1 + eq. 3 = eq. 2 so this subcase has 36.

In total, we have $169 + 133 + 36 = 338$.

Problem 23 (2022 12A/24)

We count the number of strings of length n using digits 0 through n (in base $n + 1$, for example) which satisfy this. We call the signature of each path $(r_0, r_1, \dots, r_{n-1}, r_n)$. A signature is valid if $r_0 + \dots + r_j \geq j$ for all j . Using a similar argument as we made with Catalan paths, if a signature satisfies $r_0 + r_1 + \dots + r_n = n$, exactly one of the $n + 1$ cyclic rotations of each signature is valid.

Therefore, exactly $\frac{1}{n+1}$ of the $(n + 1)^n$ strings we can make work, so the answer is $(n - 1)^{n+1}$. The answer to the problem is 9^{11} .

51 clubs; 14 hours